

Before You Scale: The Cost of Supervision Mismatch in PRM Distillation

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Abstract

Process reward models (PRMs) are often trained from automatically generated step-level supervision. However, when a small verifier trained in this way fails, it is often unclear whether the cause is student capacity, the training recipe, or the supervision itself. This distinction has direct practical consequences: when supervision itself is the bottleneck, any compute spent scaling the student or tuning the recipe is wasted until the data problem is fixed. We hit this ambiguity in practice. Privileged (answer-aware) labeling improves our teacher on MATH (+0.051 first-error F1), yet a 1.5B student distilled from those labels is no better than one trained on answer-blind labels, and a positive control shows the pipeline does track coarse teacher quality. To isolate the cause, we fix the student and evaluation path, varying only the supervision. Gold labels from non-MATH source splits (GSM8K, OmniMath) transfer reliably to ProcessBench-MATH (0.75–0.77 mean ROC-AUC over four seeds per source), while OlympiadBench-source supervision is high-variance across seeds, showing format match alone is not sufficient. Generated teacher labels are markedly weaker under the same path (0.62 mean ROC-AUC). The failure is primarily one of rendering convention: re-rendering the same generated labels into the ProcessBench first-error convention raises this baseline to 0.74, recovering most of the gap to gold supervision. The practical conclusion for efficient PRM development is that a cheap source- and format-matched diagnostic, roughly 0.3 GPU-hours against the ~ 15 GPU-hours of the recipe sweep it replaces, can reveal supervision mismatch before further compute is committed to larger students or broader recipe searches.

1 Introduction

Process reward models assign scores to intermediate reasoning steps and are increasingly used to supervise or verify mathematical reasoning systems (Lightman et al., 2023; Wang et al., 2023; Zhang et al., 2025). Because human step-level annotation is expensive, many PRM pipelines rely on automatically generated labels from stronger models. This raises a practical question for efficient reasoning systems: when a distilled verifier performs poorly, should we scale the student and training budget, or should we first question whether the generated supervision matches the target evaluation distribution?

We arrive at this question from a concrete failure. In our pipeline, a labeling teacher with privileged (answer-aware) access to reference solutions produces measurably better step-error labels in a characterizable difficulty window (§3), yet a small student verifier distilled from those labels is no better than one distilled from answer-blind labels (§4). A positive control shows the training pipeline does transmit coarse teacher quality, so the null demands a diagnosis rather than a shrug.

We provide the diagnosis with a fixed-path ablation. We fix the verifier architecture, training path, and evaluation benchmark, and vary the source, annotation provenance, and format of step-level supervision. The student is a Qwen2.5-3B model with a lightweight score head. The target benchmark is full ProcessBench-MATH, evaluated over 1,000 shuffled solutions containing 6,505 reasoning steps and 594 error steps (Zheng et al., 2024). We

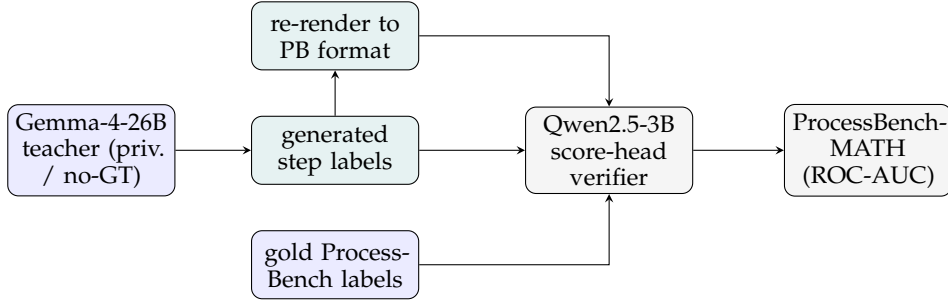


Figure 1: The controlled diagnostic. A fixed Qwen2.5-3B score-head verifier is trained under one identical path on interchangeable supervision sources—raw generated teacher labels (privileged or no-GT), the same labels re-rendered into the ProcessBench first-error format, and gold ProcessBench-style labels from non-MATH splits—and evaluated on full ProcessBench-MATH. Only the supervision source and its rendering vary; the student, optimizer, and evaluation are held fixed.

compare ProcessBench-style supervision from non-MATH source splits against generated teacher labels from the privileged/no-ground-truth labeling pipeline.

The same 3B verifier that learns a transferable ProcessBench-MATH scorer from GSM8K or OmniMath ProcessBench-style labels remains weak when trained on generated teacher labels. This contrast is not a state-of-the-art PRM claim: a public Qwen2.5-Math-7B-PRM800K baseline (Qwen Team, 2025) is stronger on the same split. Instead, the contribution is diagnostic. Under a fixed student and fixed optimizer, supervision compatibility corresponds to a large difference in transfer quality. The high-variance OlympiadBench transfer shows that source distribution still matters even when the annotation format matches. More importantly, we demonstrate via a format-rerender cell that rendering convention constitutes a major bottleneck for generated labels.

This suggests that some failed PRM distillation runs should first be treated as possible data-distribution failures rather than immediate evidence that the student is too small or recipe is wrong. For efficient PRM development, this establishes a cost asymmetry: in our setup, a single format-matched 500-step LoRA diagnostic costs approximately 0.3 GPU-hours, against approximately 15 GPU-hours for the generated-label recipe sweep it would have replaced.

In summary, we contribute (1) a characterization of when privileged supervision helps a labeling teacher, (2) a verified negative: that advantage does not distill into a small answer-blind verifier, with pipeline sensitivity confirmed by a positive control, and (3) a diagnosis: under a fixed training and evaluation path, supervision format and distribution, not student capacity, dominate the failure, with a format-rerender cell identifying rendering convention as the operative bottleneck.

2 Related Work

Process supervision and PRMs. Lightman et al. (2023) showed that process supervision substantially outperforms outcome supervision on MATH, releasing PRM800K. Because human step annotation is expensive, Math-Shepherd (Wang et al., 2023) constructs process supervision automatically and uses it for re-ranking and step-level RL. Generative verifiers such as GenPRM (Zhao et al., 2025) and ThinkPRM (Khalifa et al., 2025) reason before judging; they are close to our teacher in spirit but are deployed as inference-time verifiers rather than as teachers whose labels are distilled into a separate answer-blind student. Weak-supervision approaches train PRMs without dense human process labels (Sun et al., 2025), and noise-aware training tackles noisy process supervision directly (Xie et al., 2026); our controlled gold-label comparison is complementary because it first asks whether the current generated-label distribution supports the target verifier at all. ProcessBench frames

Tier	Benchmark (teacher)	no-GT	+answer	+solution	Δ (solution)
Easy	GSM8K (Gemma, $N = 50$)	0.889	0.856	0.837	-0.052 (n.s.)
Hard	MATH (Gemma, $N = 400$)	0.726	0.726	0.777	+0.051 [0.01, 0.09]
Hardest	OlympiadBench (Gemma, OE)	-	-	-	-0.03 (n.s.)
Hard	MATH (Qwen-27B, $N = 150$)	0.653	0.690	0.735	+0.082

Table 1: The privilege sweet spot across difficulty tiers (first-error F1), with cross-family replication. The OlympiadBench row is evaluated on the open-ended subset under a separate scoring protocol; we report only the paired within-run Δ rather than per-condition values on an incomparable scale.

earliest-error identification as a benchmark on which PRMs often fail to generalize across sources (Zheng et al., 2024), and Qwen2.5-Math PRMs provide strong public calibration points (Zhang et al., 2025; Qwen Team, 2025). Concurrent work studies PRM transfer across the same source splits: ProcessLID (Bai et al., 2026) reports explicit transfer tables, and RetrievalPRM (Zhu et al., 2025) frames PRM failures as out-of-distribution issues. Our contribution relative to this thread is the controlled contrast between generated teacher labels and ProcessBench-style gold labels under one fixed student, optimizer, and evaluation path.

Privileged information and expert-amateur contrasts. Our framing descends from CLEAR (Rufail et al., 2025) and LightReasoner (Wang et al., 2025), which extract signal from the gap between a stronger and a weaker model; we invert the axis from capacity to information, contrasting a privileged and an answer-blind view of the same teacher. The experiments relate to learning using privileged information (Vapnik and Vashist, 2009), with a boundary our results make concrete: privileged information can improve the teacher’s per-example inference without producing labels that distill into a better answer-blind student. Recent on-policy self-distillation methods condition a teacher on privileged solution information and distill token-level guidance into the same model’s policy on its own roll-outs (Zhao et al., 2026; ?). We distill off-policy step-error labels into a separate verifier, so our negative result identifies a failure mode for format-sensitive verifier supervision rather than contradicting that line; indeed our diagnosis is consistent with the distribution-matching argument motivating on-policy distillation (Li et al., 2026), observed from the verifier side. Data-centric analyses argue that systematic changes in data quality can dominate model-side changes (Mazumder et al., 2022); our result is a PRM-specific instance.

3 The Privilege Context (Teacher Level)

To generate the automated labels for our distillation study, we used a stronger labeling teacher (Gemma-4-26B), selected by an upper-bound bake-off with ground-truth access (Appendix A.1). A natural assumption in automatic supervision is that giving the teacher privileged (answer-aware) access to a reference solution will uniformly improve its labeling quality. In preliminary probes, we found this to be conditionally true: privileged access provides a measurable benefit most strongly in a mid-difficulty window where the problem is hard enough that the teacher cannot self-verify, yet tractable enough that the teacher can exploit the reference solution. On MATH, privilege adds +0.051 first-error F1 (95% CI [0.01, 0.09], $N = 400$), replicating across model families (Qwen-27B, +0.082), while the gap is not significant on GSM8K or on OlympiadBench¹ (Table 1). The privileged signal must also be substantive, supplying only the correct final answer, with no worked reasoning, flips zero teacher predictions relative to the answer-blind condition.

¹The near-zero GSM8K gap is measured on the $N = 50$ teacher probe under the first-error F1 protocol. In the larger same-source labeling run of §6 ($N = 400$, exact-match metric), the privileged teacher does show a +0.048 advantage on GSM8K. The two agree that the GSM8K gap is small relative to MATH; the sweet-spot claim concerns where the advantage is large.

Student (teacher labels)	ROC-AUC	PR-AUC	F1	Error recall
Privileged + critique	0.631	0.130	0.170	0.357
Privileged, score-only	0.624	0.135	0.184	0.427
No-GT + critique	0.641	0.121	0.198	0.467

Table 2: Step-level transfer at 1.5B ($N_{\text{train}} = 1000$, $N_{\text{eval}} = 400$). The no-GT student is highest on ROC-AUC, so privilege does not transfer.

Verifier	pass@1	PRM re-rank	Majority vote	Oracle pass@N
No-GT student	0.337	0.373	0.391	0.517
Privileged student	0.335	0.349	0.382	0.518

Table 3: Downstream best-of- N re-ranking ($N = 1000$ problems, 8 candidates, gemma-2-9b-it generator). Neither student beats majority vote.

4 Does the Teacher Advantage Transfer?

The teacher-level advantage does not distill into a small student verifier. The student is Qwen2.5-1.5B-Instruct with a score-and-critique head, trained via LoRA on teacher-produced score and critique labels ($N_{\text{train}} = 1000$, $N_{\text{eval}} = 400$). Table 2 shows the step-level result: the no-GT-trained student is highest on threshold-free ROC-AUC (0.641 vs. 0.631), so privilege does not produce a better student. Downstream, on best-of- N re-ranking over 1,000 problems with 8 candidates (Table 3), the no-GT verifier again re-ranks better than the privileged one (0.373 vs. 0.349 accuracy), and neither beats the majority-vote baseline (≈ 0.39); the oracle (0.517) shows the failure is the verifiers’, not the candidate pool’s.²

Two controls scope this null. First, it is not pipeline insensitivity: relabeling the same solutions with a deliberately weaker teacher (Qwen2.5-3B in place of Gemma-4-26B) and training an identically-configured student produces a significant degradation (+0.0497 ROC-AUC for the strong teacher, 95% CI [0.0206, 0.0783], $p = 0.0012$, paired bootstrap over 6,505 steps), so the training path does track coarse teacher quality. Second, the teacher-level and student-level orderings can invert even with source problems held fixed: in the same-source control of §6, the privileged teacher’s labels are more accurate at the teacher level (first-error exact match 0.800 vs. 0.752 on 400 GSM8K problems), yet the student trained on them is significantly weaker (0.5494 vs. 0.6183 mean ROC-AUC, $p = 0.0004$). A better labeler does not imply a better student. The teacher’s privilege advantage is real but diffuse: privileged and no-GT labels disagree on $\sim 31\%$ of steps, and the disagreement is nearly symmetric (1,001 vs. 956 steps flagged in each direction), leaving no clean directional signal for a 1.5B student to latch onto. This flat null prompted the core investigation of this paper: is the failure to distill a student-capacity problem, or a supervision problem?

5 Experimental Setup

Student and training path. All trained verifiers in §6 use the same Qwen2.5-3B-Instruct backbone with a scalar score head. We train with LoRA adapters, a score-only objective, balanced batches, binary cross-entropy for the ProcessBench-source runs, and 500 training steps. For generated-label baselines, we report the strongest generated-label recipes found in our preliminary ablations (Appendix A.4): a privileged-teacher BCE recipe and a no-ground-truth rank-loss recipe. The generated-label baselines are reported over four seeds, matching the GSM8K and OmniMath ProcessBench-source rows.

²The answer-equivalence check behind these re-ranking numbers fell back to string matching rather than symbolic verification, which understates absolute accuracy on non-canonical but equivalent MATH answers. The fallback applies identically to every column, so the comparisons are unaffected in direction; absolute values are lower bounds.

Training source	Seeds	ROC-AUC	PR-AUC	Best F_1^*	Fixed F_1	Pred. err. rate
GSM8K gold $\rightarrow$ MATH1000	0-3	0.7515 (0.7256-0.7760)	0.2188 (0.1935-0.2317)	0.3144	0.2503	0.5111
OmniMath gold $\rightarrow$ MATH1000	0-3	0.7694 (0.7539-0.7869)	0.2524 (0.2277-0.2877)	0.3276	0.3055	0.2920
OlympiadBench gold $\rightarrow$ MATH1000	0-1	0.6509 (0.5854-0.7163)	0.1619 (0.1209-0.2029)	0.2384	0.1938	0.3661
Qwen2.5-Math-7B-PRM800K (public)	–	0.8379	0.3254	0.3991	0.3953	0.1436
Generated privileged labels, BCE	0-3	0.5475 (0.5007-0.5883)	0.0966 (0.0881-0.1045)	0.1853	0.1089	0.2903
Generated no-GT labels, rank loss	0-3	0.6062 (0.5390-0.6755)	0.1254 (0.0911-0.1672)	0.2142	0.1918	0.7341
Same-source GSM8K gen. priv BCE	0-3	0.5494 (0.4680-0.6371)	0.0985 (0.0809-0.1235)	0.1931	0.1741	0.4696
Same-source GSM8K gen. no-GT BCE	0-3	0.6183 (0.5849-0.6614)	0.1152 (0.1049-0.1314)	0.2168	0.2088	0.4629
Same-source priv BCE, PB-format	0-3	0.6762 (0.5869-0.7891)	0.1761 (0.1175-0.2811)	0.2554	0.2385	0.5121
Same-source no-GT BCE, PB-format	0-3	0.7366 (0.7011-0.7555)	0.2478 (0.2266-0.2722)	0.3215	0.3120	0.1703

Table 4: Full ProcessBench-MATH ($N = 1000$) transfer results under an identical Qwen2.5-3B score-head training/eval path. Parentheses give seed ranges. PB-format rows re-render the same generated labels into the ProcessBench first-error convention. *Best F1 is threshold-swept on the eval slice; treat as diagnostic. The OlympiadBench-source row has two completed seeds (§6.3).

Evaluation. All results evaluate on the same shuffled ProcessBench-MATH split of 1,000 solutions. The split contains 6,505 total steps and 594 error steps. We use exact serial scoring with batch size 1 for headline metrics. We report ROC-AUC and PR-AUC as threshold-free measures, alongside a fixed-threshold F1 diagnostic (Appendix A.3 reports held-out calibrated F1).

Label sources. ProcessBench-source training uses the first 400 ProcessBench examples from each source configuration, flattened into step-level labels. Generated-label training uses automatically labeled MATH-style traces from Gemma-4-26B. Exact problem-string overlap between each source training file and the MATH-1000 evaluation split is zero.

6 Diagnosing the Null: Capacity or Supervision?

6.1 Format-matched labels transfer; raw generated labels remain weak

Table 4 shows the main result. With the student, training path, and evaluation held fixed, ProcessBench-style labels from GSM8K and OmniMath transfer strongly to full ProcessBench-MATH. Generated teacher labels are much weaker in their raw rendering.

Every GSM8K and OmniMath ProcessBench-source seed significantly beats both raw generated-label baselines under sequence-cluster bootstrap (resampling whole solutions rather than individual steps, which are correlated). Against the best generated-label baseline overall (the strongest single generated-label checkpoint from the preliminary sweep: no-GT rank loss, seed 0, ROC-AUC 0.6324), the ROC-AUC gaps range from +0.093 to +0.155 across the eight GSM8K/OmniMath seeds (two-sided $p = 0.0010$; Appendix A.2). In the same-source GSM8K control, even the weakest GSM8K gold seed beats the best same-source generated no-GT seed by +0.064 ROC-AUC (95% CI [0.045, 0.083], $p = 0.0004$), so the generated-label weakness is not explained by source distribution alone.³

6.2 The format-rerender mechanism

The raw generated-label failure is not a student capacity ceiling. The follow-up format-rerender cell (Table 4, bottom rows) changes the diagnosis: using the exact same Gemma-4 generated labels but re-rendering them to match the ProcessBench first-error convention raises same-source no-GT to 0.7366 mean ROC-AUC and same-source privileged to 0.6762. The no-GT and privileged 4-seed mean-score artifacts reach 0.7599 and 0.7789 ROC-AUC

³Run-to-run variance on the generated-label track is substantial and we disclose it rather than aggregate it away: re-running the privileged BCE seed-0 configuration from scratch produced 0.477 ROC-AUC against a canonical 0.550, with both runs collapsing to a predicted-error rate of 0. The seed ranges in Table 4 reflect genuine training instability on generated labels, not measurement noise; the gold-label and PB-format rows show no comparable degeneracy.

respectively, statistically tied with the GSM8K gold mean-score artifact under sequence-cluster bootstrap. This shows that rendering convention, not raw student capacity or even raw teacher label quality, constitutes the major bottleneck for generated labels in this distillation setup.

6.3 The source distribution still matters

This is not a blanket claim that any non-MATH process labels transfer cleanly to MATH. OlympiadBench-source training is high-variance in our runs: two seeds give 0.585 and 0.716 ROC-AUC on full MATH-1000, one significantly below the best generated-label baseline (-0.047 , $p = 0.0010$) and the other significantly above it ($+0.084$, $p = 0.0010$). We treat OlympiadBench as a boundary diagnostic rather than a third clean headline source; the pattern still supports a source-distribution view, but the source split matters.

6.4 The failure is not explained by the score-head machinery alone

A plausible concern is that the 3B score-head architecture is simply too weak to learn step-level mathematical verification. Our diagnostics argue against this as the primary explanation. Training the same score-head path on a matched ProcessBench-MATH split (first 200 shuffled MATH examples, evaluated on the next 200) reaches 0.7129 ROC-AUC. Frozen-representation probes tell the same story: class-balanced logistic probes on frozen Qwen2.5-3B step-boundary embeddings trained on generated teacher labels transfer weakly (0.52–0.56 ROC-AUC), while the matched ProcessBench-gold probe reaches 0.631. Training-recipe changes did not rescue the generated labels either: a ~ 10 -configuration sweep over score-plus-critique distillation, pure ranking loss, BCE-plus-rank, and longer BCE training left privileged-label transfer weak or negative, often favoring the no-ground-truth labels instead (Appendix A.4). This suggests that the failure is not merely that the student representation cannot expose the target; the supervision distribution and format are doing the heavy lifting.

7 Limitations

This study is diagnostic rather than prescriptive. We evaluate one main student architecture and score-head training path, so a different student family or objective may narrow the gap. The gold-vs-generated contrast varies provenance, source-problem distribution, and label format simultaneously; the same-source GSM8K control removes the source-distribution confound for one source split, and the format-rerender cell shows that rendering convention explains much of the raw failure, but label content and provenance remain coupled, since the re-rendered teacher first-error position still disagrees with gold on roughly a quarter of source solutions.

Generated-label training is markedly less stable than gold-label training (seed ranges span ~ 0.09 – 0.14 ROC-AUC vs. ~ 0.03 – 0.05 for gold), and some runs collapse to a degenerate all-negative score head. We read this instability as part of the finding rather than noise. The OlympiadBench-source row has two seeds and is treated as a boundary diagnostic, not a headline source. The positive control’s weak-teacher student is also miscalibrated (near-zero error-prediction rate), so part of that measured gap may reflect outright degeneracy rather than a smooth teacher-quality gradient. The 1.5B and 3B tracks differ in loss recipe and evaluation N , so the observation that generated-label transfer stays weak at both scales is suggestive, not a clean capacity ablation. Downstream validation of the 3B result is incomplete: we do not show the gold-supervised 3B verifier beats majority vote on best-of- N re-ranking. Finally, the public Qwen2.5-Math-7B-PRM800K baseline is stronger than any single gold-source seed on the same split, reinforcing that our contribution is diagnostic, not a SOTA PRM claim.

8 Conclusion

Privileged process supervision helps a labeling teacher only in a tractability window, and that advantage does not survive distillation into a small answer-blind verifier, despite a training pipeline demonstrably sensitive to coarse teacher quality. Under a fixed 3B verifier, training path, and ProcessBench-MATH evaluation, compatible ProcessBench-style supervision transfers much better than raw generated teacher labels, and re-rendering the same generated labels into the target convention recovers most of the gap. The result reframes a failed distillation attempt as a supervision-source and format mismatch diagnosis.

For efficient PRM development, this suggests a practical ordering: before scaling students or re-running the same distillation loop, test whether the supervision format itself can support the target verifier. In our setup, one 500-step LoRA diagnostic required approximately 0.3 GPU-hours, whereas the preliminary generated-label recipe sweep used approximately 15 GPU-hours. A practical takeaway is to first run a small diagnostic with format-matched labels before investing in a larger student or a new training recipe.

A Supplementary Tables

All numbers trace to committed artifacts in the anonymized supplementary material.

A.1 Teacher selection

Teacher	F1	Correct acc	Error acc	Parse fail
Gemma-4-26B (primary)	0.909	1.000	0.833	0.000
Qwen-27B	0.873	0.917	0.833	0.000
Qwen-35B-A3B	0.839	1.000	0.722	0.019
Qwen-4B-Thinking	0.000	1.000	0.000	0.846

Table 5: Teacher bake-off, with-GT ceiling ($N = 30$). A reasoning-tuned candidate collapses under the structured-output protocol.

A.2 Sequence-cluster bootstrap gaps

Model A	Model B	ROC-AUC gap	95% CI (p)
GSM8K gold seed 0	best generated baseline	+0.111	[0.087, 0.134] (0.0010)
OmniMath gold seed 0	best generated baseline	+0.121	[0.096, 0.146] (0.0010)
GSM8K gold seed 0	generated privileged BCE	+0.193	[0.164, 0.221] (0.0010)
OmniMath gold seed 0	generated privileged BCE	+0.203	[0.176, 0.232] (0.0010)
Qwen PRM800K	OmniMath gold seed 3 (best)	+0.051	[0.033, 0.068] (0.0010)
PB-format no-GT 4-seed mean	raw no-GT 4-seed mean	+0.126	[0.104, 0.148] (0.0004)
PB-format priv 4-seed mean	raw priv 4-seed mean	+0.207	[0.183, 0.231] (0.0004)
GSM8K gold 4-seed mean	PB-format priv 4-seed mean	+0.004	[-0.008, 0.017] (0.4983)

Table 6: Representative sequence-cluster bootstrap ROC-AUC gaps. Full pairwise tables are in the anonymized supplementary material.

A.3 Held-out threshold calibration

Model	Calib. F_1	ROC-AUC	PR-AUC	Pred. err. rate
GSM8K gold seed 0	0.3166	0.7452	0.2339	0.2488
OmniMath gold seed 0	0.3062	0.7483	0.2292	0.2742
Qwen2.5-Math-7B-PRM800K	0.3855	0.8372	0.3242	0.1091
Generated privileged BCE	0.1673	0.5490	0.1002	0.3212
Best generated-label baseline	0.2085	0.6314	0.1431	0.1695

Table 7: Held-out threshold calibration (threshold on first 200 MATH sequences, evaluated on remaining 800).

A.4 Generated-label training-recipe ablations

Generated-label recipe	Priv ROC-AUC	No-GT ROC-AUC	Note
Score+critique MSE, early run	0.4582	0.4870	collapsed near all-negative
Score-only MSE	0.4844	0.4776	stable but weak
Feedback-token truncation fixed	0.4514	0.5093	stable, negative for priv.
BCE, error weight 3	0.5411	0.4700	best short priv. ablation
Rank-only, balanced batches	0.4939	0.6306	no-GT wins strongly
BCE+rank, balanced batches	0.5214	0.6140	no-GT wins strongly
BCE, error weight 3, longer	0.4782	0.4971	longer BCE degrades signal

Table 8: Preliminary generated-label ablations on the Qwen2.5-3B score-head path. Representative rows from a ~ 10 -configuration sweep; no variant rescues privileged-label transfer.

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